

Chemistry Playbook

Introduction · Elements & Formulae · Matter · Mixtures & Separation · Air · Water

ICSE Class 6 · Full Book · Final Exam Ready

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Chapter names and order can differ slightly between schools and publishers. Follow your school's exam syllabus.

How to use this playbook

Chapter	Topic
1	Introduction to Chemistry (and lab apparatus)
2	Elements, Compounds, Symbols and Formulae
3	Matter: Physical and Chemical Changes
4	Pure Substances, Mixtures and Separation
5	Air and Its Constituents
6	Water

The chapter numbers in your school's book may be in a different order. Teach by topic.

⭐ **Exam Tip: How to score full marks in Chemistry**

1. Learn the **symbols and valencies** by heart. They are used everywhere.
2. Write formulae carefully. **Capital letter first, small letter second** (Co is cobalt, but CO is carbon monoxide!).
3. For separation questions, always give the **principle** (why the method works).
4. Draw apparatus diagrams with labels.

1. Introduction to Chemistry

- **Chemistry** is the branch of science that studies **matter**: what it is made of (**composition**), its **properties** and the **changes** it goes through.
- **Antoine Lavoisier** is called the **Father of Modern Chemistry**.
- **Ancient India:** Acharya **Nagarjuna** worked with metals and medicines. **Kanad** proposed the idea of tiny particles called "**parmanu**" (atoms).

Importance of chemistry

Field	Contribution
Food	Fertilisers, pesticides, preservatives
Health	Medicines, antiseptics, vaccines, soaps
Clothes	Synthetic fibres (nylon, polyester), dyes
Home	Detergents, cooking gas, plastics, cement
Industry	Fuels, metals, glass, paper

The dark side of chemistry

- Pollution of air, water and soil. Plastic waste. Harmful pesticides. Explosives and chemical weapons.
- We must use chemicals **wisely and safely**.

Laboratory apparatus

Apparatus	Use
Test tube	Heating or mixing small amounts of substances
Beaker	Holding and mixing liquids
Conical (Erlenmeyer) flask	Mixing liquids without spilling
Round-bottom flask	Heating liquids evenly
Funnel	Pouring liquids into narrow openings and filtering
Measuring cylinder	Measuring the volume of liquids
Burner (Bunsen burner or spirit lamp)	Heating
Tripod stand and wire gauze	Supporting apparatus while heating
Test tube holder	Holding a hot test tube
China dish	Evaporating liquids
Watch glass	Covering a beaker, or evaporating small amounts
Glass rod	Stirring
Tongs	Holding hot objects

Lab safety rules

1. Never taste or directly smell chemicals.
2. Point the mouth of a test tube **away** from yourself and others while heating.
3. Wear an apron and goggles, and tie long hair back.
4. Never add water to acid. **Add acid to water** slowly.
5. Wash your hands after the experiment.

🧠 **Memory Trick**

"Do what you oughta, add acid to water!" Kanad = "Kan" (particle): the Indian sage who thought about atoms.

🎲 **Play It in Class: Apparatus Charades**

One student acts out using an apparatus (stirring with a glass rod, holding a hot test tube...). The others guess its name and its use. Then show the real apparatus or a picture.

✏️ **Quick Check**

1. Define chemistry.
2. Who is called the Father of Modern Chemistry?
3. Name the apparatus used to (a) measure the volume of a liquid (b) evaporate a solution.
4. Write two lab safety rules.

2. Elements, Compounds, Symbols and Formulae

- An **element** is a pure substance made of **only one kind of atom**. It cannot be broken into simpler substances. For example: iron, oxygen, gold. There are **118** known elements.
- A **compound** forms when **two or more elements combine chemically in a fixed ratio**. For example: water (H_2O), common salt (NaCl), carbon dioxide (CO_2).
- **Atom:** the smallest particle of an element. **Molecule:** the smallest particle of a substance that can exist on its own (O_2, H_2O).

Kinds of elements

Metals	Non-metals	Metalloids
Shiny (lustrous)	Dull (except iodine)	Some properties of both
Good conductors of heat and electricity	Poor conductors (except graphite)	
Malleable (beaten into sheets) and ductile (drawn into wires)	Brittle	
Solid (except **mercury**, a liquid)	Solid, liquid (**bromine**) or gas	
Iron, copper, gold, aluminium	Oxygen, carbon, sulphur, hydrogen	Silicon, boron, arsenic

Symbols of elements

Element	Symbol	Element	Symbol
Hydrogen	H	Sodium (Natrium)	**Na**
Helium	He	Potassium (Kalium)	**K**
Carbon	C	Iron (Ferrum)	**Fe**
Nitrogen	N	Copper (Cuprum)	**Cu**
Oxygen	O	Silver (Argentum)	**Ag**
Sulphur	S	Gold (Aurum)	**Au**
Chlorine	Cl	Lead (Plumbum)	**Pb**
Calcium	Ca	Mercury (Hydrargyrum)	**Hg**
Magnesium	Mg	Tin (Stannum)	**Sn**
Aluminium	Al	Zinc	Zn
Phosphorus	P	Iodine	I

- A symbol is **one capital letter**, or **one capital + one small letter**.
- Some symbols come from **Latin names** (shown in brackets above).

Valency

- **Valency** is the combining capacity of an element (how many "hands" it has to join with others).

Valency 1	Valency 2	Valency 3
H, Na, K, Ag, Cl	O, Mg, Ca, Zn, Cu (also 1), Fe (also 3)	Al, N, Fe (also 2)

Radicals (groups of atoms that act as one unit):

Radical	Formula	Valency
Hydroxide	OH	1
Nitrate	NO_3	1
Ammonium	NH_4	1
Carbonate	CO_3	2
Sulphate	SO_4	2

Writing formulae: the criss-cross method

1. Write the symbols side by side (the metal first).
2. Write each valency above its symbol.
3. **Criss-cross** the valencies and write them as subscripts.
4. Simplify if possible, and don't write the subscript 1.

Example: Aluminium oxide: Al has valency 3 and O has valency 2 → **Al_2O_3** **Example:** Calcium chloride: Ca has valency 2 and Cl has valency 1 → **$CaCl_2$** **Example:** Magnesium oxide: Mg has valency 2 and O has valency 2 → Mg_2O_2 → simplify → **MgO**

Common formulae

Compound	Formula
Water	H_2O
Common salt (sodium chloride)	NaCl
Carbon dioxide	CO_2
Ammonia	NH_3
Methane	CH_4
Calcium carbonate (chalk, limestone)	$CaCO_3$
Sodium hydroxide (caustic soda)	NaOH
Sulphuric acid	H_2SO_4
Hydrochloric acid	HCl

🧠 Memory Trick

Latin symbols, a story to remember: "**Na**ughty **K**id **Fe**ll in **Cu**p of **Ag**ra **Au**nty, **Pb** (plumber) **Hg** (hugged) **Sn** (snake)." Na = sodium, K = potassium, Fe = iron, Cu = copper, Ag = silver, Au = gold, Pb = lead, Hg = mercury, Sn = tin. **Criss-cross = "swap your hands"**: the valency of one becomes the subscript of the other.

🎲 Play It in Class: Valency Handshake

Each student wears a card with an element and holds up as many hands as its valency (H shows 1 hand, O shows 2, Al shows 3). Students must "hold all hands" to form a compound. H + H + O = H_2O! Who can make Al_2O_3?

✏️ Quick Check

1. Define element and compound with one example each.
2. Write the symbols: sodium, iron, silver, lead.
3. Write the formula of: (a) magnesium chloride (b) calcium oxide (c) sodium sulphate.
4. Which metal is a liquid at room temperature?

3. Matter: Physical and Chemical Changes

- **Matter** is anything that has mass and occupies space. Its three states are **solid, liquid and gas**.
- **Interconversion:** heating changes solid → liquid → gas, and cooling reverses it.

Physical change vs chemical change

Physical change	Chemical change
No new substance forms	A **new substance** forms
Usually **reversible**	Usually **irreversible**
Only the shape, size or state changes	The composition changes
Little or no energy change	Heat, light or sound is often given out or taken in
Melting ice, dissolving sugar, tearing paper, magnetising iron	Burning paper, rusting iron, cooking food, curdling milk, ripening fruit

Signs of a chemical change

A change of colour, gas bubbles, a smell, heat or light, or a solid forming (precipitate).

Rusting

- Iron + oxygen + moisture → **rust** (iron oxide, reddish-brown)
- **Prevent it by** painting, oiling or greasing, or **galvanising** (coating with zinc).

Desirable and undesirable changes

- **Desirable:** cooking food, ripening fruit, making curd
- **Undesirable:** rusting, food going bad, milk going sour by accident

🧠 **Memory Trick**

Physical = "Fake makeover": it looks different, but it's still the same thing inside. **Chemical = "Real transformation"**: a brand-new substance, and you usually can't go back. (You can't un-cook a vada pav!)

📍 **Mumbai Connection**

Iron railings and the gates of buildings near the sea in Mumbai **rust faster**. Salty, moist sea air speeds up rusting. That's why they are painted again every year before the monsoon!

🎲 **Play It in Class: Physical or Chemical? Stand-Up Game**

The teacher calls out a change ("Lighting a diya!", "Melting butter!", "Making curd!"). Students stand up for chemical and stay seated for physical. Anyone who's wrong explains the correct answer.

✏️ Quick Check

1. Give two differences between physical and chemical changes.
2. Classify: (a) dissolving salt in water (b) burning a candle's wick (c) rusting.
3. What three things are needed for rusting?
4. Give two ways to prevent rusting.

4. Pure Substances, Mixtures and Separation

- A **pure substance** has a fixed composition and fixed properties: elements and compounds.
- A **mixture** has two or more substances mixed in **any ratio**, with **no chemical combination**. Each part keeps its own properties.

Mixture	Compound
Parts mixed in **any proportion**	Elements combined in a **fixed proportion**
Parts keep their own properties	The properties are new and different
Can be separated by **physical** methods	Can be separated only by **chemical** methods
Example: air, salt water, soil	Example: water, salt, CO_2

- **Homogeneous mixture:** the same throughout (salt in water, air)
- **Heterogeneous mixture:** not uniform, so you can see the parts (sand in water, oil in water)

Methods of separation

Method	Principle (why it works)	Example
Hand picking	Different size and colour	Stones from rice and dal
Winnowing	Light husk is blown away by wind, while heavy grain falls	Husk from grain
Threshing	Beating to separate grain from the stalks	Wheat
Sieving	Different particle sizes	Flour from bran, sand at construction sites
Magnetic separation	One part is magnetic	Iron filings from sand
Sedimentation and decantation	The heavy solid settles, then the liquid is poured off	Muddy water
Filtration	Insoluble solid particles can't pass through filter paper	Sand from water, tea leaves
Evaporation	The liquid turns into vapour, leaving the dissolved solid	Salt from seawater
Sublimation	One part changes directly from solid to gas	Camphor or ammonium chloride from salt
Distillation	Liquids with different boiling points (or a solid from a liquid, keeping the liquid)	Pure water from salt water
Separating funnel	Liquids that don't mix (immiscible), with different densities	Oil and water
Churning	Lighter fat separates when spun	Butter from curd or milk

- **Filtrate:** the clear liquid that passes through the filter. **Residue:** the solid left on the filter paper.

🧠 Memory Trick

Your kitchen is a chemistry lab! Tea strainer = filtration. Chhalni (sieve) = sieving. Making butter (makhan) = churning. Picking stones from dal = hand picking.

📍 Mumbai Connection

Salt pans near **Bhandup, Wadala and Uran** make salt by **evaporating seawater** in the sun. The water evaporates and salt crystals are left behind!

🎲 Play It in Class: Separation Challenge

Give each team a mixture in a cup: sand + iron filings + salt + water. Teams plan the order of separation (magnet → filter → evaporate) and present it. The best plan with the correct principles wins.

✏️ Quick Check

1. Give two differences between a mixture and a compound.
2. Name the method of separation: (a) husk from grain (b) iron pins from sand (c) camphor from salt.
3. What are the filtrate and the residue?
4. How is salt obtained from seawater?

5. Air and Its Constituents

- **Air** is a **mixture** of gases. It surrounds the Earth as the **atmosphere**.

Composition of air (by volume)

Gas	Percentage
Nitrogen	about 78%
Oxygen	about 21%
Argon and other noble gases	about 0.9%
Carbon dioxide	about 0.03–0.04%
Water vapour, dust, smoke	Varies

Why is air a mixture?

1. Its composition varies from place to place.
2. Its gases can be separated by physical methods (such as the fractional distillation of liquid air).
3. Each gas keeps its own properties.
4. No heat or light is given out when the gases are mixed.

Importance of each gas

Gas	Use
Oxygen	Needed for **breathing (respiration)** and **burning (combustion)**, and used in hospitals and by divers
Nitrogen	Slows down burning, plants need it to make proteins, used to make fertilisers and to keep food fresh in chip packets
Carbon dioxide	Plants need it for **photosynthesis**, used in fire extinguishers and fizzy drinks
Water vapour	Forms clouds and rain
Noble gases	Argon is used in bulbs and neon in advertisement signs

Experiments with air

- **Air has mass:** a balloon filled with air weighs more than an empty one.
- **Air occupies space:** push an upside-down glass into water. The water doesn't fill the glass, because air is inside.
- **Oxygen is needed for burning:** cover a burning candle with a glass jar. It goes out once the oxygen is used up.
- **Lime water test:** carbon dioxide turns lime water **milky**.

Air pollution

- **Causes:** vehicle smoke, factory smoke, burning garbage, firecrackers, construction dust
- **Effects:** breathing problems (asthma), acid rain, global warming
- **Prevention:** public transport, CNG vehicles, planting trees, no burning of garbage

🧠 **Memory Trick**

"78 + 21 = 99, Nitrogen and Oxygen are the big two!" Nitrogen is the "calm friend" who slows down fire. Oxygen is the "excited friend" who helps fire burn!

📍 **Mumbai Connection**

Mumbai's **AQI (Air Quality Index)** gets worse in winter, around Diwali and near construction sites. Students can check the AQI on a phone app for one week and make a chart. Which day was worst, and why?

🎲 **Play It in Class: Candle Under the Jar**

Light 3 candles of the same size. Cover one with a small glass, one with a big glass, and leave one open. Which goes out first? (The small glass, because it has the least oxygen.) Students write down their observation and conclusion.

✏️ **Quick Check**

1. Name the two main gases in air and their percentages.
2. Give two reasons why air is a mixture.
3. Which gas turns lime water milky?
4. Write two uses of nitrogen.

6. Water

- Water covers about **71%** of the Earth's surface. About **97%** of it is salty seawater. Only about **3%** is fresh water, and most of that is frozen in ice caps and glaciers.
- Water is a **compound** of hydrogen and oxygen (**H_2O**), with hydrogen and oxygen in the ratio **2 : 1 by volume**.

Sources of water

- **Rain** (the purest natural form), rivers, lakes, ponds, wells, springs, seas and oceans, groundwater

Water is the "universal solvent"

- Water dissolves more substances than any other liquid.
- **Solute:** the substance that dissolves (sugar). **Solvent:** the liquid it dissolves in (water). **Solution** = solute + solvent.
- **Saturated solution:** no more solute can dissolve at that temperature. Heating usually lets more solute dissolve.

Water cycle

Evaporation (seas, rivers) → condensation (forming clouds) → precipitation (rain) → collection (rivers, groundwater) → evaporation again

Purifying water (from a city supply)

Screening/sedimentation → loading (adding alum) → filtration (sand beds) → chlorination (kills germs) → storage and supply

Purifying water at home

Boiling, filtering, adding chlorine tablets, alum, RO/UV water purifiers

Hard and soft water

- **Soft water** lathers easily with soap (rainwater).
- **Hard water** doesn't lather easily, because it contains dissolved calcium and magnesium salts.

Saving water

Close taps, fix leaks, use a bucket instead of a shower, **harvest rainwater**, reuse RO waste water for mopping.

> 🧠 **Memory Trick**
> **Water cycle = "E-C-P-C": "Every Cloud Pours Carefully"**: Evaporation, Condensation, Precipitation, Collection.

📍 Mumbai Connection

Mumbai's drinking water comes from lakes outside the city, such as **Vaitarna, Tansa, Bhatsa and Vihar**. The BMC treats it at plants such as **Bhandup**, one of Asia's biggest water treatment plants. In years with poor rain, the BMC announces **water cuts**. That's why saving water matters!

🎲 Play It in Class: Water Detective

Each student counts how many litres of water their family uses in one day (one bucket ≈ 15 L). The class adds up all the totals, then plans 5 ways to save 20%.

✏️ Quick Check

1. What percentage of the Earth's water is fresh water?
2. Why is water called a universal solvent?
3. Write the steps of the water cycle in order.
4. What is hard water?

Full-Book Revision Test

Section A (1 mark each)

1. Write the symbol for gold.
2. Name a liquid non-metal.
3. Give the formula of common salt.
4. What percentage of air is oxygen?
5. Name the method used to separate oil from water.
6. Is the curdling of milk a physical or a chemical change?

Section B (2–3 marks each)

7. Write the formulae: aluminium chloride, calcium carbonate, sodium hydroxide.
8. Give three differences between metals and non-metals.
9. Explain why rusting is a chemical change.
10. How would you separate a mixture of sand, salt and iron filings?
11. Describe an experiment to show that air occupies space.
12. Name the steps in purifying water for a city supply.

Section C (4 marks each)

13. Distinguish between mixtures and compounds (four points).
14. Write the uses of oxygen, nitrogen and carbon dioxide.
15. Explain the water cycle with a labelled diagram.

Answer Key

Ch 1: (1) Chemistry is the branch of science that studies matter, its composition, its properties and the changes it goes through. (2) Antoine Lavoisier (3) (a) Measuring cylinder (b) China dish (4) Never taste chemicals, point test tubes away from people while heating, add acid to water (any two)

Ch 2: (1) Element: one kind of atom (iron). Compound: elements chemically combined in a fixed ratio (water). (2) Na, Fe, Ag, Pb (3) (a) $MgCl_2$ (b) CaO (c) Na_2SO_4 (4) Mercury

Ch 3: (1) A physical change makes no new substance and is reversible. A chemical change makes a new substance and is usually irreversible. (2) (a) Physical (b) Chemical (c) Chemical (3) Iron, oxygen (air) and moisture (water) (4) Painting, oiling or galvanising

Ch 4: (1) In a mixture the parts are in any ratio and keep their properties. In a compound the ratio is fixed and the properties are new. (2) (a) Winnowing (b) Magnetic separation (c) Sublimation (3) The filtrate is the clear liquid that passes through. The residue is the solid left on the filter paper. (4) By evaporating seawater in salt pans using the Sun's heat.

Ch 5: (1) Nitrogen about 78%, oxygen about 21% (2) Its composition varies, its gases can be separated physically, and each gas keeps its own properties (any two) (3) Carbon dioxide (4) It slows down burning, it is used to make fertilisers, and it is used in food packets (any two)

Ch 6: (1) About 3% (2) Because it dissolves more substances than any other liquid. (3) Evaporation → condensation → precipitation → collection (4) Water that doesn't lather easily with soap, because it contains calcium and magnesium salts.

Revision test: (1) Au (2) Bromine (3) NaCl (4) About 21% (5) A separating funnel (6) Chemical (7) $AlCl_3$, $CaCO_3$, NaOH (8) Metals are lustrous, conduct electricity and are malleable. Non-metals are dull, don't conduct electricity and are brittle. (9) A new substance (iron oxide) forms, and the change cannot be reversed. (10) Use a magnet to remove the iron filings. Dissolve the rest in water and filter to remove the sand. Evaporate the filtrate to get the salt. (11) Push an upside-down glass into water. The water doesn't rise inside, because air fills the space. (12) Sedimentation → loading with alum → filtration → chlorination → supply (13) Proportion (any vs fixed), properties (kept vs new), separation (physical vs chemical), energy (no change vs heat or light given out or taken in) (14) Oxygen: breathing and burning. Nitrogen: fertilisers and food packing. Carbon dioxide: photosynthesis and fire extinguishers. (15) Evaporation from water bodies → vapour cools and condenses into clouds → precipitation as rain → collection in rivers, lakes and the ground → repeat